Hirschsprung's disease, also known as congenital aganglionosis, is characterized by a tonic contracture of the bowel starting at the dentate line of the anus and extending proximally for a variable distance. This contracture produces a functional bowel obstruction. As a result, the more proximal bowel dilates dramatically. Symptoms range from mild constipation to toxic enterocolitis requiring an emergency colostomy.



Laparoscopic 'pull-through' procedures are often performed to excise the contracted portion of the bowel and re-anastomose it to the rectum. An older, more traditional approach involves a left lower quadrant incision through which a loop colostomy is performed and then the procedure completed when the child is older.

Patient positioning for the procedure will either be supine or lithotomy, depending upon the surgeon's preference. Blood loss is usually mild (<5 mL/kg). The patient population is often from newborn to 18 months of age with males being affected 4 times more than females. The patients may exhibit varying degrees of malnourishment, but are often otherwise healthy. Approximately 5% of patients with Hirschsprung's disease have Trisomy 21. Epidural anesthesia is typically acceptable for pain management, but will not provide the degree of muscle relaxation needed to perform a corrective procedure.

Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 1331-1334.

Copyright 2015 Prodigy Anesthesia, Inc. It is unlawful to copy, distribute, and/or sell this material.